# Homework 6
## Combinatorics
## Fall 2026

**Due:** Thursday, October 8, 2026 23:59
**Submit:** Gradescope, through Blackboard's assignment area.
Please write your homework by hand. Start each exercise on a new page, scan your work to a legible PDF, and clearly mark the pages associated with every exercise. Show the algebra used to obtain each generating function or closed formula.

**1.** (From a generating function to a recurrence)

**Points:** 10
The ordinary generating function of a sequence $(b_n)_{n\geq 0}$ is

$$B(x) = \sum_{n\geq 0} b_n x^n = \frac{1+2x+3x^2}{1-2x-x^3}.$$

(A) **4 points.** Determine initial data $b_0$, $b_1$, and $b_2$. Show your calculations. Explain which terms you disregard.

(B) **6 points.** Determine the recurrence satisfied by $b_n$, including its valid index range, and use it to compute $b_3$ and $b_4$. Explain your derivation.

**2.** (From a recurrence to a generating function)

**Points:** 10
Let $(d_n)_{n\geq 0}$ be defined by

$$d_0 = 0, \qquad d_n = 2d_{n-1} + n \quad \text{for } n \geq 1.$$

Let $D(x) = \sum_{n\geq 0} d_n x^n$.

(A) **10 points.** Multiply the recurrence by $x^n$ and sum over all $n \geq 1$. Rewrite every sum in terms of $D(x)$ and the standard series

$$\sum_{n=0}^{\infty} nx^n = \frac{x}{(1-x)^2}.$$

Solve the resulting equation for $D(x)$ and give your answer as a single rational function.

**3.** (Solving a homogeneous recurrence)

**Points:** 10
Solve the recurrence

$$u_n = 4u_{n-1} + 4u_{n-2} \quad \text{for } n \geq 2, \qquad u_0 = 2, \quad u_1 = 7.$$

Your solution must include the characteristic polynomial, its roots, the general form of a solution to the recurrence, and the use of the two initial conditions to determine the constants. Check your closed formula at $n = 2$ against a direct recurrence calculation.

**4.** (A nonhomogeneous recurrence with two driving terms)

**Points:** 10
Solve the recurrence

$$a_n = 5a_{n-1} - 6a_{n-2} + 2^n + 2n - 7 \quad \text{for } n \geq 2, \qquad a_0 = 3, \quad a_1 = 4.$$

(A) **4 points.** Treat $2^n$ and $2n - 7$ as separate driving terms. For each driving term, write an appropriate trial form for a particular solution. Explain why the trial for $2^n$ needs an extra factor of $n$.

(B) **4 points.** Substitute both trial forms into the recurrence and determine their unknown coefficients. Add the two particular solutions to the general homogeneous solution.

(C) **2 points.** Write down the general formula for the homogeneous solution.

(D) **4 points.** Use the initial conditions to determine the remaining constants. Check your final formula at $n = 2$ against a direct recurrence calculation.